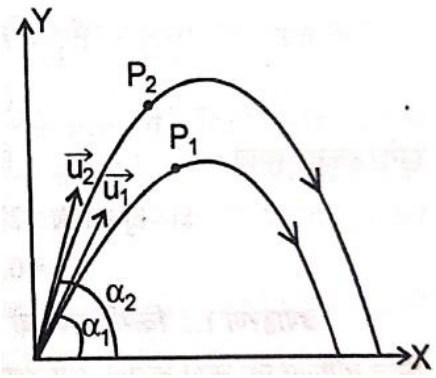
AS1108: Applied Physics – Fall 2026
Problem Set 1: Classical Mechanics

1. Show that the motion of the projectile as seen from another projectile will always be a straight line.



2. Initial positions of the particles are $(4\hat{\imath} - 4\hat{\jmath} + 7\hat{k})$ and $(2\hat{\imath} + 2\hat{\jmath} + 5\hat{k})$ respectively. If the velocity of the first particle is 0.4 $(\hat{\imath} - \hat{\jmath} + \hat{k})$ m/s. Find the velocity of the particle so that they can collide after 10 seconds?

3. Consider a particle moving in a straight line with a velocity $v = (0.5t^2 + t)\ m/s$. Find the position and acceleration at t=4 s. It is given that at t=0; x=0.

4. Consider a ball thrown vertically upward with a velocity of 15 m/s from a height of 24 m above the ground. Determine the maximum height reached by the ball and corresponding time. (Neglect air resistance and take g=9.8 m/s^2.)

5. If the position vector of a particle $\vec{r} = \hat{\imath}cos\omega t + \hat{\jmath}sin\omega t$ show that its velocity is perpendicular to $\vec{r}$.

6. A body of mass 100 grams is moving with a velocity$(2\hat{\imath} + 3\hat{\jmath} + 7\hat{k}) \times 10^2$m/s in a reference frame which itself is rotating with an angular velocity $(\hat{\imath} - \hat{\jmath}) \times 10^{-2}$rad/s with respect to another stationary frame. Determine the magnitude of the Coriolis force acting on the body.

7. The distance between the atoms of oxygen (m_o=16 units) and carbon (m_c=12 units) in a CO molecule is 1.12 Å. Determine the position of centre of mass of CO molecule from carbon atom.

8. The position of two particles of mass 100 and 300 grams are $(2\hat{\imath} + 5\hat{\jmath} + 13\hat{k})\ cm$ and $(-6\hat{\imath} + 4\hat{\jmath} - 2\hat{k})\ cm$. These particles are moving with velocities $(10\hat{\imath} - 7\hat{\jmath} - 3\hat{k})\ cm/s$ and $(7\hat{\imath} - 9\hat{\jmath} + 6\hat{k})\ cm$/s respectively. Calculate (i) position vector of the centre of mass, (ii) velocity of second particle with respect to centre of mass.

9. Centre of mass of three particles having masses 10, 20, 30 grams are at (1,-2,3). Find the position of fourth particle of mass 40 grams, so the centre of mass of all the particles is at (1,1,1).

10. A bomb moving with a velocity $\left(40\hat{\imath} + 50\hat{\jmath} - 25\hat{k}\right) m/s$ divides into two pieces. If the mass ratio of these two pieces is 1:4 and the smaller piece moves with a velocity $\left(200\hat{\imath} + 70\hat{\jmath} + 15\hat{k}\right) m/s$, then determine the velocity of bigger piece. Also what will be the velocities of these pieces from centre of mass frame.

11. Three particles each of masses 10, 20, and 40 grams are moving with velocity 20 cm/s along X, Y and Z axes and collide. After collision third particle come to rest and second particle moves with a velocity $\left(10\hat{\jmath} + 5\hat{k}\right) cm$/s. Determine the velocity of first particle.

12. A stone at rest divided into three pieces in explosion. First piece of mass 1 kg moves with a velocity 12 m/s. Second piece of mass 2 kg moves with a velocity 8 m/s in a direction perpendicular to the direction of first piece. If the third piece moves with a velocity 40 m/s, then determine its mass.

13. The position vector of a particle of mass 100 grams at any moment is $\left(\vec{r} = 7\hat{\imath} + 3\hat{\jmath} + \hat{k}\right) m$ and its velocity is $\left(\vec{v} = -3\hat{\imath} + \hat{\jmath} + 5\hat{k}\right) m/s$. Determine the angular momentum of a particle about an origin (0,0,0).

14. A force $\left(\vec{F} = 3\hat{\imath} + 2\hat{\jmath} - 4\hat{k}\right)$ N is acting on a particle whose position vector about origin is $\left(\vec{r} = 2\hat{\imath} - 4\hat{\jmath} + 2\hat{k}\right) m$. Determine torque about origin and its magnitude.

15. A stone of mass 100 grams tied to a thread of length 50 cm is moving in a circular path at the rate of 2 rev./s. Determine its angular momentum.

16. Particles of mass 100 and 200 grams are separated by a massless rod of length 30 cm. The centre of mass of the system is at the origin of the reference frame. Rod is in XY-plane and makes an angle 30 degree with the Y-axis. Determine the inertia coefficients I_{xx} and I_{xy}.

17. Three particles of same mass m are placed at (a,0,0), (0,a,0) and (0,0,a). Determine all inertia coefficients of the system.